Department of Physics
PCS 125 – Physics: Mechanics

Insert in the circle the first initial of your family name

Please Print Clearly:

Student:_____ _____
Last Name *First Name*

Student Number: _____ **Student Section:** _____

Please check the box for your instructor

☐ **Dr. Springer** ☐ **Dr. Toronov** ☐ **Dr. Xu** ☐ **Dr. Abbas** ☐ **Dr. Beauchemin**

Final Exam – April 26, 2018

Version A– Please bubble question #50 with A

Read the following attentively

- **Ensure that your name and student number are correct in the bubble sheet.**
- **Duration of the exam is 180 minutes.**
- **The exam consists of 25 multiple-choice questions. Answer ALL questions and show all work.**
- **Each multiple-choice question is worth 1 mark**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer. If no work or correct argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.**
  - **Transfer your choices to the bubble sheet only when you are sure of them.**
  - **Some answers may be rounded off.**
- **Pagers and cell phones must be silenced and placed out of reach. Hats and earphones are not allowed. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.**
- **Sharing of pencils, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **The calculator models allowed for the test are either Sharp EL-546 or Casio FX-991 (or similar models). If you are found to have another type of calculator, it may be confiscated for the duration of the test.**
- **The exam booklet has a total of 18 pages. You are responsible for checking that you have all pages. Several blank pages are included at the end of the exam which you may use for calculations**

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Sign here to signify you read the instructions

[This page is intentionally blank and can be used for calculations]

1.) A 0.28kg block on a horizontal frictionless surface is attached to an ideal massless spring whose spring constant is 500 N/m. The block is pulled from its equilibrium position at x=0.0m to a displacement of $x = +0.80$m and is released from rest. The block then executes simple harmonic motion along the horizontal x-axis. When the displacement is $x = -0.052$m, what is the acceleration of the block?

a.) 93 $\mathrm{m/s^2}$
b.) 47 $\mathrm{m/s^2}$
c.) 69 $\mathrm{m/s^2}$
d.) 230 $\mathrm{m/s^2}$
e.) 280 $\mathrm{m/s^2}$

2.) In this problem, times are denoted hh:mm (h = hours, m=minutes). Treat Earth as a uniform sphere with radius 6371 km and mass 5.97×10^{24}kg. Treat Mars as a uniform sphere with radius 3390 km and mass 6.39×10^{23} kg.

A pendulum clock keeps perfect time when it is located on the surface of the Earth. An astronaut then bring the clock to the surface of Mars, but makes no other changes to the clock. The astronaut's ship's computer is perfectly synchronized with clocks on Earth by satellite. When his ship's computer reads exactly 12:00pm he sets the pendulum clock to 12:00pm and starts it running. One hour later, when the ship computer reads 1:00pm, what time does the pendulum clock read?

a.) 1:23 pm (The pendulum clock is 23 minutes fast)
b.) 1:38 pm (The pendulum clock is 38 minutes fast)
c.) 1:00 pm (The pendulum clock still keeps perfect time)
d.) 12:22 pm (The pendulum clock is 38 minutes slow)
e.) 12:37 pm (The pendulum clock is 23 minutes slow)

3.) A sinusoidal wave of wavelength 1.50 m and amplitude 0.20 m travels on a string with a speed of 3.00 m/s in the $-x$ direction. At $t = 0$s, the piece of string at $x = 0$m has a displacement of 0.0m and is moving downwards in the $-y$ direction. Which of these equations is the correct representation for this wave?

a.) y(x,t) = (0.20 m) sin[(4/3)π x + 4π t + π]
b.) y(x,t) = (0.20 m) sin[(4/3)π x + 4π t]
c.) y(x,t) = (0.20 m) sin[(2/3)π x + 4π t]
d.) y(x,t) = (0.20 m) sin[(2/3)π x + 4π t + π]
e.) y(x,t) = (0.20 m) sin[(4/3)π x + 2π t + π]

4.) A traveling sinusoidal wave on a string is described by the following equation $y(x,t) = (1.9\text{m})\sin(5x - 2t)$ where t is in seconds and x in metres. If the string has a linear mass density of 0.01 kg/m , what is the tension in the string?

a.) 2.5×10^{-2} N
b.) 4.0×10^{-3} N
c.) 1.6×10^{-3} N
d.) 6.3×10^{-2}N
e.) 3.3×10^{-2}N

5.) A loudspeaker on a tall pole broadcasts sound waves equally in all directions. What is the speaker's power output if the sound level is 90 dB at a distance of 20m?

a.) 11 W
b.) 5.0 W
c.) 3.5 W
d.) 24 W
e.) 0.8 W

6.) A bat locates insects by emitting ultrasonic "chirps" and then listening for echoes from the bugs. Suppose a bat chirp has a frequency of 25,000 Hz, which is outside the range of human hearing (human hearing range is 20 Hz – 20,000 Hz) . What is the *minimum* speed at which the bat would need to fly in order for you (a stationary observer) to barely hear the chirp? Also determine the direction the bat must fly. You may assume the speed of sound is 343m/s.

a.) 38 m/s away from you
b.) 68 m/s away from you
c.) 68 m/s toward you
d.) 86 m/s toward you
e.) 86 m/s away from you

7.) Two harmonic waves travelling in opposite directions interfere to produce a standing wave described by $y(x,t) = (3\text{m})\sin(2\,x)\cos(5t)$ where x is measured in metres and t is measured in seconds. What is the wavelength of the interfering waves?

a. 3.14 m
b. 1.00 m
c. 6.28 m
d. 12.0 m
e. 2.00 m

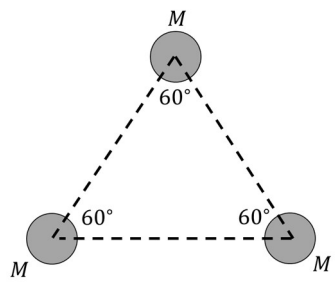
8.) Consider a hypothetical dense astronomical object with a mass of 2×10^{30}kg (about the mass of the sun), and a radius of 1.4×10^{14}m. Assuming you could stand on the surface of this object, if you fired a gun straight into the air, how high above the surface would the bullet go before turning around? The mass of a bullet is 5.0g and its speed is 800 m/s when it leaves the gun. Ignore any frictional or drag forces on the bullet.

a.) 3.3×10^{4} m
b.) 7.1×10^{13} m
c.) 2.1×10^{14} m
d.) 4.2×10^{14} m
e.) Infinitely high (it escapes from the surface and never turns around).

9.) Which of the following frequencies could **NOT** be present as a standing wave in a 1.0m long pipe open at one end and closed at the other. The fundamental frequency for this pipe is 85 Hz.

a.) 85 Hz.
b.) 170 Hz.
c.) 255 Hz.
d.) 425 Hz.
e.) 595 Hz.

10.) Three identical spheres with mass 50-kg are held at the corners of an equilateral triangle 30cm on each side. These spheres are in outer space, far from the influence of any other stars or planets. If one of the spheres is released what will be the magnitude of the initial acceleration of that sphere immediately after the release?

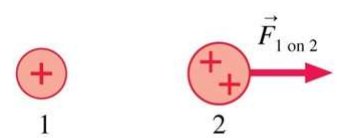


a.) 3.7×10^{-8} m/s^2
b.) 2.5×10^{-8}m/s^2
c.) 1.9×10^{-8} m/s^2
d.) 4.2×10^{-8}m/s^2
e.) 6.4×10^{-8} m/s^2

11.) An electron is in an electric field of uniform magnitude and direction $\vec{E} = (0.2 \times 10^{-7}\,\hat{\imath} + 0.7 \times 10^{-7}\hat{\jmath})\frac{N}{C}$. If the electron's velocity is 3.1×10^4 m/s in the positive x direction at time $t = 0$s, what will be its ***speed*** 3.0s later?

a.) 2.05×10^4 m/s
b.) 3.69×10^4 m/s
c.) 4.22×10^4 m/s
d.) 6.38×10^4 m/s
e.) 7.36×10^3 m/s

12.) Two positively charged spheres are placed near each other as shown in the figure. Sphere 1 has half as much charge on it as Sphere 2. The magnitude of the force exerted on sphere 2 by sphere 1 is $\vec{F}_{1\text{ on }2} = +20\text{N}\,\hat{\boldsymbol{\imath}}$. [The $+x$ axis points to the right]. What is the force felt by sphere 1?

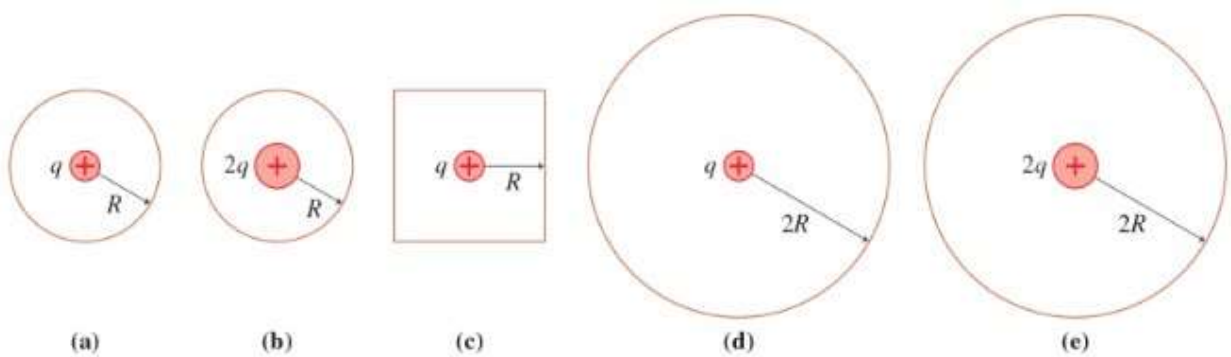


a.) $+10\text{N}\,\hat{\boldsymbol{\imath}}$
b.) $+20\text{N}\,\hat{\boldsymbol{\imath}}$
c.) $-10\text{N}\,\hat{\boldsymbol{\imath}}$
d.) $-20\text{N}\,\hat{\boldsymbol{\imath}}$
e.) $-40\text{N}\,\hat{\boldsymbol{\imath}}$

13.) A 3.0 cm **diameter** circle lies in the xz-plane in a region where the electric field is $\vec{E} = (2500\hat{\imath} + 1500\hat{\jmath} - 4000\hat{k})$N/C. What is the magnitude of the electric flux through the circle?

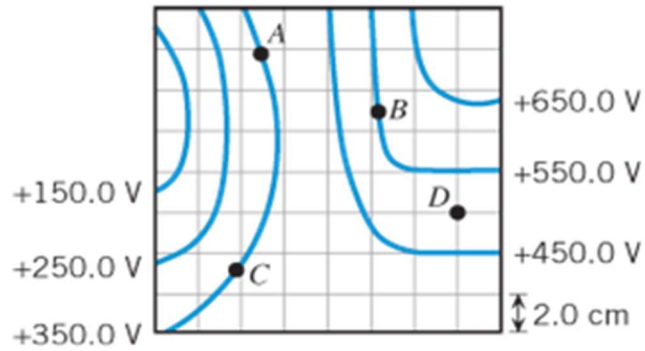
a.) 0 N· m^2 /C
b.) 1.06 N·m^2 /C
c.) 1.77 N·m^2 /C
d.) 2.83 N·m^2 /C
e.) 3.33 N·m^2 /C

14.) The figure shows five three-dimensional closed shapes surrounding positive charges: four spheres and a cube. Rank in order, from largest to smallest, the electric fluxes Φ_a to Φ_e through surfaces (a) to (e).



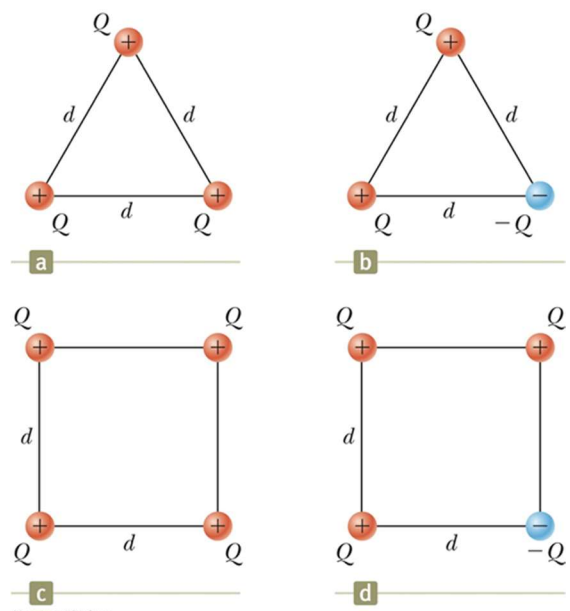
a.) $\Phi_a > \Phi_b > \Phi_d > \Phi_c = \Phi_e$
b.) $\Phi_b = \Phi_a = \Phi_d > \Phi_c = \Phi_e$
c.) $\Phi_e > \Phi_c = \Phi_b > \Phi_b = \Phi_a$
d.) $\Phi_b = \Phi_e > \Phi_a = \Phi_c = \Phi_d$
e.) $\Phi_b > \Phi_a = \Phi_d = \Phi_c = \Phi_e$

15.) The drawing shows a graph of a set of equipotential surfaces seen in cross section. Each is labeled according to its electric potential. A $+2.8\times10^{-7}$ C point charge is placed at position *A*. Find the work that is done on the point charge by the electric field when it is moved from *A* to *B*. You may assume the charge is at rest at both places.



a.) $+2.5\times10^{-5}$ J
b.) -3.6×10^{-5} J
c.) -5.6×10^{-5} J
d.) $+4.2\times10^{-7}$ J
e.) -7.6×10^{-7} J

16.) Rank the electric potential energies of the systems of charges shown in the figure below from largest to smallest.



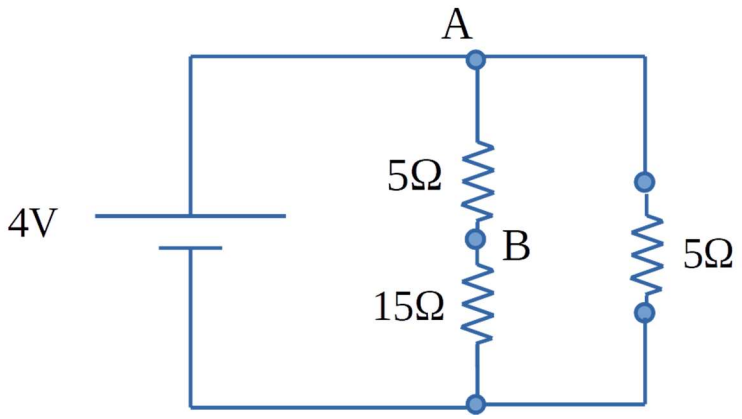
a.) c > d > a > b
b.) d > c = b > a
c.) c = d > a = b
d.) c > a > d > b
e.) d > a > b > c

17.) A light bulb is rated at 30 W when operated at 120 V. How much charge passes through the light bulb in 1.0 min?

a.) 17 C
b.) 15 C
c.) 14 C
d.) 13 C
e.) 60 C

18.) A voltmeter is connected between points A and B in the circuit shown. What does it read?

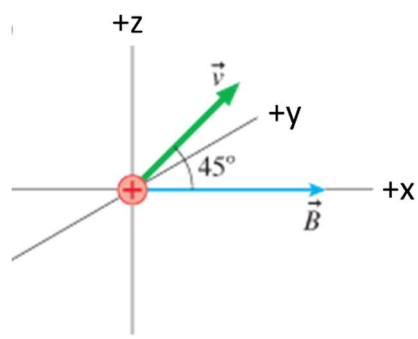
a.) 0.0V since no current flows in this branch
b.) 1.0V
c.) 2.0V
d.) 3.0V
e.) 4.0V



19.) A particle with charge $+q$, mass m and speed v is observed traveling in a circular path of radius R in a region of space where a uniform magnetic field exists. A second particle is identical to the first one except has a charge $+2q$, and travels in the same region of space at the same speed. What would be radius of the second particle's circular path?

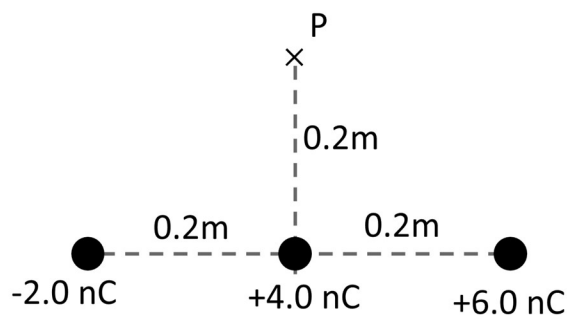
a.) $8R$
b.) $4R$
c.) $2R$
d.) $R/2$
e.) $R/4$

20.) A proton moves in the magnetic field $\boldsymbol{B}$ =0.50 $\hat{\imath}$ T with a speed of 1.0×10^{7} m/s in the xz-plane as shown. The angle shown is the angle that the proton's velocity makes with the $+x$ axis. What is magnetic force **F** on the proton?



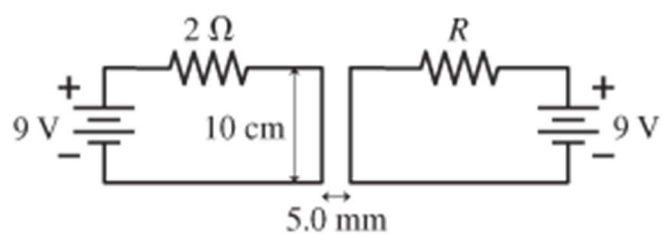
a.) $+6.1 \times 10^{-11}\hat{\imath}$ N
b.) $+8.0 \times 10^{-13}\hat{\jmath}$ N
c.) $-8.0 \times 10^{-13}\hat{\jmath}$ N
d.) $-5.7 \times 10^{-13}\hat{\jmath}$ N
e.) $+5.7 \times 10^{-13}\hat{\jmath}$ N

21.) Three point charges are placed along the x axis as shown ($1\text{nC} = 10^{-9}\text{C}$). What is the electric potential at the point P? [Assume the usual convention that the electric potential is equal to zero infinitely far away from the charges].



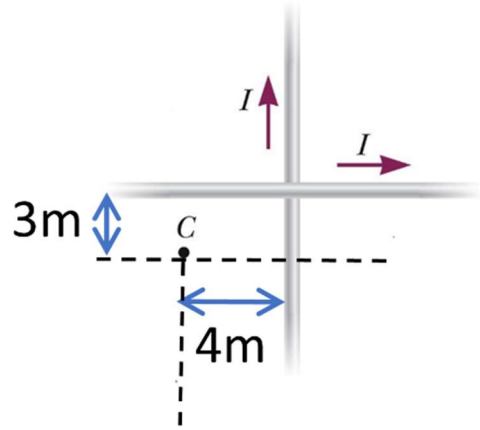
a.) -307V
b.) $+307\text{V}$
c.) $+154\text{V}$
d.) -360V
e.) $+360\text{V}$

22.) Two 10-cm-long parallel wires shown are separated by 5.0 mm. For what value of the resistor R will the force that these two wires exert on each other be $5.4 \times 10^{-5}\text{N}$? [Because the wires have a length much larger than their separation, you may treat them as "infinite" wires].



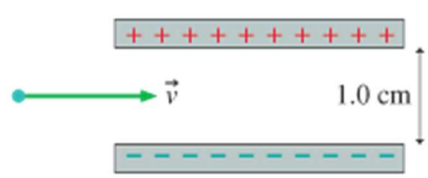
a.) 15 Ω
b.) 3.0 Ω
c.) 51 Ω
d.) 0.7 Ω
e.) 100 Ω

23.) Two infinitely long wires **each** carry 10A of current as shown. The point C is 3.00m below the horizontal wire, and 4.00m to the left of the vertical wire. What is the magnitude and direction of the magnetic field at point C?



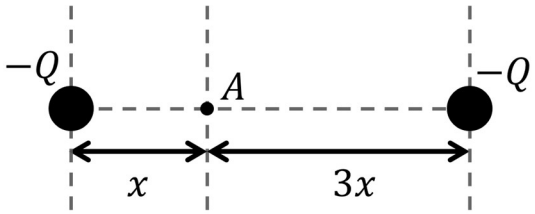
a.) 1.67×10^{-7}A (into the page)
b.) 1.67×10^{-7}A (out of the page)
c.) 1.17×10^{-6}A (into the page)
d.) 1.17×10^{-6}A (out of the page)
e.) 4.00×10^{-7}A (out of the page)

24.) An electron travels with speed 1.0×10^{7} m/s between the two parallel charged plates as shown. The plates are separated by 1.0 cm and are charged by a 200 V battery. What magnetic field strength and direction will allow the electron to pass between the plates without being deflected?



a.) 1.0 T into the page
b.) 3.5 T upward
c.) 0.5 T downward
d.) 0.15 T out of the page
e.) 0.002 T into the page

25.) Two identical negative charges -Q are a distance $4x$ apart. What are the magnitude and direction of the electric field at point A, a distance x from the left charge along the line connecting the charges?



a.) $|\vec{E}| = 2\,k_eQ/3x^2$ (pointing to the right)
b.) $|\vec{E}| = k_eQ/3x^2$ (pointing to the right)
c.) $|\vec{E}| = 2\,k_eQ/9x^2$ (pointing to the right)
d.) $|\vec{E}| = k_eQ/9x^2$ (pointing to the left)
e.) $|\vec{E}| = 8\,k_eQ/9x^2$ (pointing to the left)

<u>This is version A.</u>
<u>Please bubble in Question #50 with A</u>

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